

TRIAI x Google DeepMind's AI Research foundations course

Challenge 2: Data Card

Project title: Early Detection of Toxic Language through Ubuntu-Aligned Latent Probing of LLMs

Participants of the project:

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Datasets to be used	What kind of datasets are to be used? Synthetic or real? What are the sources of these datasets?	In this project both real datasets as well as synthetic dataset will be used. Real datasets intended to be used include but not limited to: HateXPlain: A Benchmark Dataset for Explainable Hate Speech Detection by Mathew et al. (2021). AfriHate: A Multilingual collection of Hate Speech and Abusive Language datasets for African Languages by Shamsudeen et al. (2025). ToxicGen: A Large-Scale Machine-Generated Dataset for Adversarial and Implicit Hate Speech Detection by Hartvigsen et al. (2022). In addition to these real datasets, we have also generated synthetic dataset that reflects ubuntu's value of relationality (Birhanu, S. (2026). AI-generated English-Amharic synthetic dataset for toxicity detection).
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Input vs Output	• What are users inputting? • What are the outputs or labels your model will predict or generate?	The inputs will be the texts of users in the LLMs, and the outputs will be the classification of whether these texts are: labels (safe/ hate speech; ጥሩ ንግግር/ የጥላቻ ነግግር), relational harm (yes or no; አዎ አለ/ የለም), relational target (group or individual; ቡድን/ ግለሰብ), and severity (high/ low/ zero; ከፍተኛ/ ዝቅተኛ/ ዜሮ).
Classification Problem	• What is the classification task? • What categories or classes are you working with, and why?	Our classifications are: 1. Labels (safe or hate speech). We believe that identifying labels such as whether the inputs are hate speech or not is important in order to prevent further harms in the society. 2. Relational harm (yes or No). Since our focus is on African society particularly based on the value of Ubuntu's relationality, understanding whether the inputs of users harm relations of the society or not is very important. 3. Relational target (Group or Individual).

		After identifying whether there is relational harm or not, our next step will be who is targeted? Understanding whether the target is on the group or the individual is important so that further relational harm can be detected easily and can be avoided. 4. Severity. Our last job will be understanding the extent of the harm. Is it high? Is it low? Or nothing?
Labeling	Where do your labels come from – domain experts, community members, crowd, AI generated sources?	The labels are from both domain experts as well as AI-generated sources.
Data creation	- How will you collect your data? - Will it come from direct sources, or will you synthesize data in low resource settings	The data comes from secondary sources (other scholarly works) as well as we have synthetically generated.
Web collection ethics	- Do you plan to source data from web? - If yes, which site or platforms? - What tools will you use? - Have you addressed issues of consent, copyright, and privacy	Yes we plan to get data from platforms such as Github and hugging face. We plan to use AI tools. We acknowledge the works of others and we will use appropriate citations, and we will not expose the privacy of individuals as much as possible (one of our main data is synthetic which avoids

		privacy problems for example)
Bias and representation	• Who might be left out of your dataset? • How will you ensure that the dataset reflects the diversity of the community and include underrepresented voices?	This project is specifically aimed at representing the neglected part of the society in toxicity detection mechanisms. Accordingly, it will be more African focus so that African values can be represented in the toxicity detection methods of AI systems. However, neither it does favour Africans exclusively nor it does exclude non-Africans. It is just to mean representing the unrepresented.
Values and community alignment	• How does your dataset design reflect the values you identified in course 1? • How will your community benefit from this dataset and how will you ensure accountability to them?	This project is aligned to Ubuntu's value of relationality that mirrors the standpoint of many Africans (the importance of relations, community, reciprocity, and interdependence). This project will be shaped by ubuntu's philosophical and ethical perspective. Respect, brotherhood/sisterhood, relational responsibility, reciprocity, and inclusiveness are among the most important values that will influence this project.
Evaluation metrics to be used	Which evaluation metrics will be used in this project	Evaluations metrics such as Accuracy, precision, Recall,

		and F1-score will be used in this project.
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